

Project Coversheet

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Project Title	Week1: Customer Sign-Up Behavior & Data Quality Audit

Instructions:

Students must download this cover sheet, use it as the first page of their project, and then save the entire document as a PDF before submission.

Project Guidelines and Rules

1. Formatting and Submission

- Format: Use a readable font (e.g., Arial/Times New Roman), size 12, 1.5 line spacing.
- Title: Include Week and Title (Example - Week 1: Travel Ease Case Study.)
- File Format: Submit as PDF or Word file
- Page Limit: 4–5 pages, including the title and references.

2. Answer Requirements

- Word Count: Each answer should be within 100–150 words; Maximum 800–1,200 words.
- Clarity: Write concise, structured answers with key points.
- Tone: Use formal, professional language.

3. Content Rules

- Answer all questions thoroughly, referencing case study concepts.
- Use examples where possible (e.g., risk assessment techniques).
- Break complex answers into bullet points or lists.

4. Plagiarism Policy

- Submit original work; no copy-pasting.
- Cite external material in a consistent format (e.g., APA, MLA).

5. Evaluation Criteria

- Understanding: Clear grasp of business analysis principles.
- Application: Effective use of concepts like cost-benefit analysis and Agile/Waterfall.

- Clarity: Logical, well-structured responses.
- Creativity: Innovative problem-solving and examples.
- Completeness: Answer all questions within the word limit.

6. Deadlines and Late Submissions

- Deadline: Submit on time; trainees who fail to submit the project will miss the “Certificate of Excellence”

7. Additional Resources

- Refer to lecture notes and recommended readings.
- Contact the instructor or peers for clarifications before the deadline.

Introduction

Rapid Scale is a fast-growing SaaS company offering tiered subscription plans designed to meet diverse customer needs. The company leverages data-driven insights through its Business Intelligence team, which analyzes user behavior to support the Monthly Business Review and optimize business strategies. By auditing customer sign-up data and understanding acquisition trends, Rapid Scale empowers its Marketing and Onboarding teams to enhance campaigns, improve engagement, and tailor offerings to customer demographics.

Data Cleaning Summary

The dataset underwent thorough cleaning to ensure accuracy and consistency across customer-related columns.

- **Duplicates** were minimal, with only one repeated email address remained the same because that doesn't affect the overall data.
- **Missing values** were addressed (**34.9%** missing data overall): 7 names, 32 emails, 7 sources, 28 regions, 6 plan selections, 8 marketing opt-ins, 10 ages, and 6 genders were noted, a total of **104 missing entries out of 298 records**.
- **Standardizations** were applied to categorical fields to ensure consistency, including capitalizing the first letter of source, region, and gender, and unifying inconsistent plan entries (e.g., "PRO" → "Pro").
- **Invalid entries**, such as "not a date" in signup date, were corrected.
- **Age values and gender categories** were checked for irregularities, with unusual entries cleaned for reliable analysis.

Key Findings & Trends

- **Marketing opt-in rates are generally low**; the majority of users do not want to receive marketing communications. The younger age groups, such as 18–25 and 26–35, also show very low opt-in rates, whereas middle-aged users (35–50) are slightly more receptive.

- **Premium and Pro are the most engaging plans**, with Premium being the most chosen in general, while Pro users tend to create the most support activity. Middle-aged users, aged between 26–50, make up the biggest proportion of Premium subscribers.
- **Data quality is an issue**, with about 35% of all entries having missing or inconsistent information. Invalid dates, duplicated emails, inconsistent gender categories, and incomplete demographic details reflect the need for improved processes in data collection.

Business Question Answers

1. Which source brought the most users last month?

Based on the data collected last month, registered users came from Instagram (4), YouTube (2), Referral (2), Facebook (1), and LinkedIn (1). Among all the sources, Instagram brought in the highest number of users. This means Instagram was the most effective source of new users during that period.

2. Which region shows incomplete or missing data?

The number of entries with an unknown region is 30, and the least amount of data was collected from the Central region. This suggests that more incomplete or missing data is likely associated with the Central region.

3. Are older users more or less likely to opt in to marketing?

The data shows users grouped by age and whether they opted into marketing. Younger users (18–25) have 29 opted-in versus 44 not, while 25–35 has 35 opted-in versus 50 not. In 35–50, opt-in equals 35, slightly higher than 34 not opted-in. For 50–70, 18 opted-in versus 23 not.

Older users (50–70) are slightly less likely to opt in compared to middle-aged (35–50), while younger users (18–25) show lower opt-in rates. Overall, middle-aged users (35–50) appear most receptive, suggesting a moderate increase in marketing interest with age, then a slight decline among the oldest group.

4. Which plan is most commonly selected, and by which age group?

The data indicates that the Premium plan is the most popular, chosen by 90 users. Breaking it down by age:

- 18–25: 21 users
- 26–35: 29 users
- 36–50: 30 users

- 51–70: 10 users

The plan is most favored by middle-aged users (26–50), accounting for 59 out of 90 users, suggesting that individuals in this age range are more likely to invest in premium services. Younger users (18–25) and older users (51–70) are less represented, indicating lower adoption in these groups. Overall, premium appeal peaks in the 26–50 age range.

5. Which plan’s users are most likely to contact support?

Based on the data, the Pro plan has the highest number of users at 47, making it the plan most likely to generate support contacts. Users on Pro may require more guidance or assistance due to advanced features compared to Basic or Premium. The next highest is Basic (38), followed by Premium (26).

Recommendations

- Improve form validation
- Fix gender and plan dropdowns
- Reduce missing region/email entries
- Target marketing campaigns using source/age insights

Data Issues or Risks

One significant data quality issue is the presence of invalid and inconsistent entries in the signup date column, including values such as “not a date”. This creates inaccuracies in time-based reporting, user journey analysis, and monthly acquisition trends. To fix this at the source, the signup form should enforce strict date validation, allowing only properly formatted timestamps.